

1 Gain law

(bars denote normalized quantities: lengths by R ; $a_o := \Delta t / (\gamma_N R)$ [1/pN] and $a_{\text{nom}} := \Delta t / \gamma_N = a_o R$ [$\mu\text{m}/\text{pN}$]; $\bar{a} := a / a_{\text{nom}}$; forces by a_o : $\bar{f} := a_o f$, so that $\bar{a} \bar{f} = \Delta \bar{w}$; $0 \leq \bar{a} \leq 1$, far field 1, contact 0; hats sit on the normalized quantities $\hat{a}_w, \hat{b}, \delta \hat{w}; e_\theta = \theta - \hat{\theta}$)

$$\begin{aligned}\bar{a}(\bar{w}) &= 1 - \frac{1}{b(\bar{w} - \bar{w}_s)} \\ \bar{a}'(\bar{a}, b) &:= \frac{d\bar{a}}{d\bar{w}} = b(1 - \bar{a})^2 \\ \bar{a}''(\bar{a}, b) &:= \frac{d\bar{a}'}{d\bar{w}} = -2b^2(1 - \bar{a})^3 \\ a_w &= a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R}\end{aligned}$$

2 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2; \quad b_{\text{true}}[k] := b_{\text{true}}(\bar{w}[k]))$$

$$\begin{aligned}\bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\ \bar{w}_d[k + 1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\ \delta \bar{w}_1[k + 1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k + 1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k + 1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ \Delta \bar{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\ \bar{a}_w[k + 1] &= \bar{a}_w[k] + b_{\text{true}}[k] (1 - \bar{a}_w[k])^2 \Delta \bar{w}[k] \\ b_{\text{true}}[k + 1] &= b_{\text{true}}[k] + b'_{\text{true}}[k] \Delta \bar{w}[k]\end{aligned}$$

$$\begin{aligned}F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i] \\ \varepsilon_w[k] &= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] - (1 - \lambda_c) n_w[k]\end{aligned}$$

3 Estimator

$$\mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b]^\top \quad (d = 2)$$

$$\begin{aligned}
\delta \hat{w}_1[k+1] &= \delta \hat{w}_2[k] \\
\delta \hat{w}_2[k+1] &= \delta \hat{w}_3[k] \\
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] \\
\Delta \hat{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \\
\hat{a}_w[k+1] &= \hat{a}_w[k] + \hat{b}[k] (1 - \hat{a}_w[k])^2 \Delta \hat{w}[k] \\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}$$

4 Error dynamics

(True – Estimator; coefficients at the estimate; $e_b[k] := b_{\text{true}}[k] - \hat{b}[k]$)

$$\begin{aligned}
e_{\Delta \bar{w}}[k] &= \Delta \bar{w}[k] - \Delta \hat{w}[k] = (1 - \lambda_c) e_{\delta \bar{w}_3}[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
b_{\text{true}}[k] (1 - \bar{a}_w[k])^2 &= \hat{b}[k] (1 - \hat{a}_w[k])^2 - 2 \hat{b}[k] (1 - \hat{a}_w[k]) e_{\bar{a}_w}[k] + (1 - \hat{a}_w[k])^2 e_b[k] + \mathcal{O}(e^2)
\end{aligned}$$

$$\begin{aligned}
e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k] \\
e_{\delta \bar{w}_2}[k+1] &= e_{\delta \bar{w}_3}[k] \\
e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \left[1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} - 2 \hat{b}(1 - \hat{a}_w) (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) \right] e_{\bar{a}_w}[k] \\
&\quad + (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 e_{\delta \bar{w}_3}[k] + (1 - \hat{a}_w)^2 (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) e_b[k] + \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w[k] \\
e_b[k+1] &= e_b[k] + b'_{\text{true}}[k] \Delta \bar{w}[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 & 1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} & (1 - \hat{a}_w)^2 \\ & & & -2 \hat{b}(1 - \hat{a}_w) (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) & \times (\Delta \bar{w}_d + (1 - \lambda_c) \delta \hat{w}_3) \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \left[0, 0, -\varepsilon_w, \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w, 0 \right]^\top$$